Getting Started with the ECU

Notice: RM Control Systems products are intended for competition, racing, and other lawful off-road applications and are not intended to be used to bypass, defeat, disable, or render inoperative any emissions-control device or system. These products are not intended for installation or use on emissions-controlled motor vehicles where such installation or use would violate applicable federal, state, or local law. The purchaser and installer are responsible for determining and maintaining compliance with all laws and regulations applicable to the vehicle and its use.

1. Download and install the STM32 Virtual COM Port (VCP) drivers:
    a. Option 1:
    From the official STMicroelectronics website (registration may be required):
    https://www.st.com/en/development-tools/stsw-stm32102.html



Get Software

Part Number	General Description	Latest version	ECCN (EU)	ECCN (US)	Supplier	Download
STSW-STM32102	STM32 Virtual COM Port Driver	1.5.0	NEC	3D991	ST	Get latest

    b. Option 2:
    From the RM Control Systems Google Drive:
    https://drive.google.com/file/d/1kvkKc6qwiY2arnYQB2QzxJUCPKBwo-Ut/view?usp=sharing

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2. Download and install tuning software:
    a. Option 1: Download and install TunerStudio MS from EFI Analytics.
    This guide will focus on using TunerStudio.
    https://www.efianalytics.com/TunerStudio/download/

    Note: There is a free version of TunerStudio with limited capabilities, or one-time pay registered versions with greater capabilities. The free version is sufficient for basic tuning, while the registered version unlocks advanced features such as VE Analyze Live, enhanced data logging, and additional tuning tools. Registered TunerStudio is recommended.

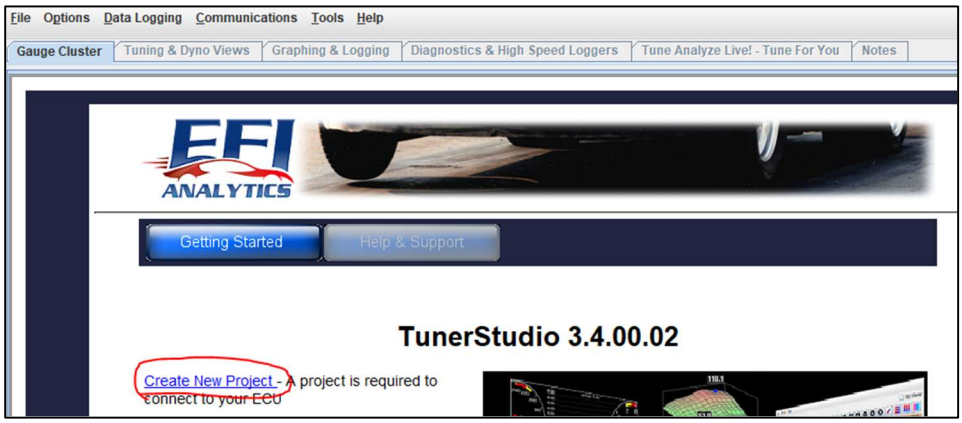
    b. Option 2: Download and install EpicTuner.
    EpicTuner is a free alternative to TunerStudio.
    https://github.com/AKCore/epictuner_public/releases

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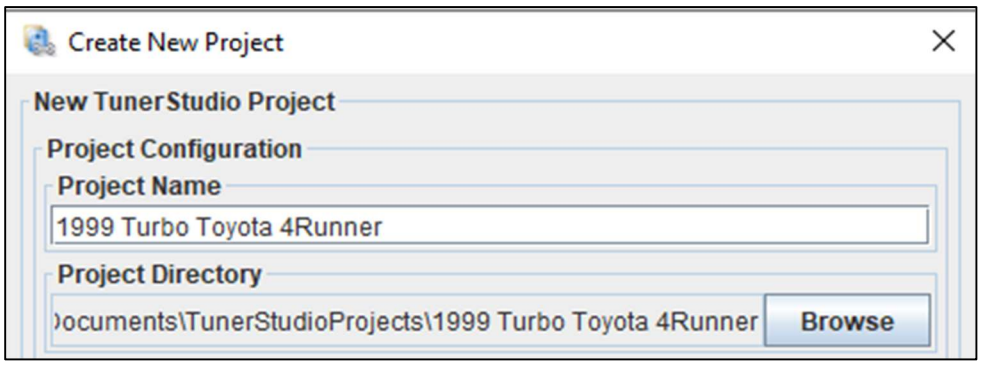
3. Create a New Project in TunerStudio:
    a. **Important:** Many USB-C cables are charging-only and cannot transfer data. Make sure you have a USB-C cable that supports data transfer (not just charging!)

    b. Plug your laptop into the ECU via the USB-C cable. Your laptop should make an audible noise acknowledging that it sees the ECU.

    c. In TunerStudio, select “Create New Project”.

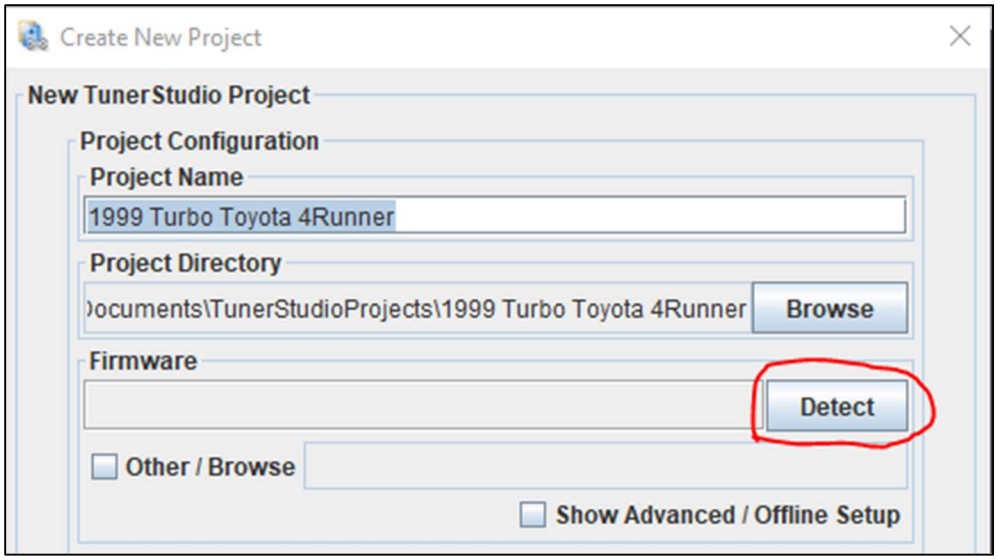


    d. Under “Project Name” choose a name for your project.
    e. Under “Project Directory” choose a location to save your project files (the default location is typically fine).

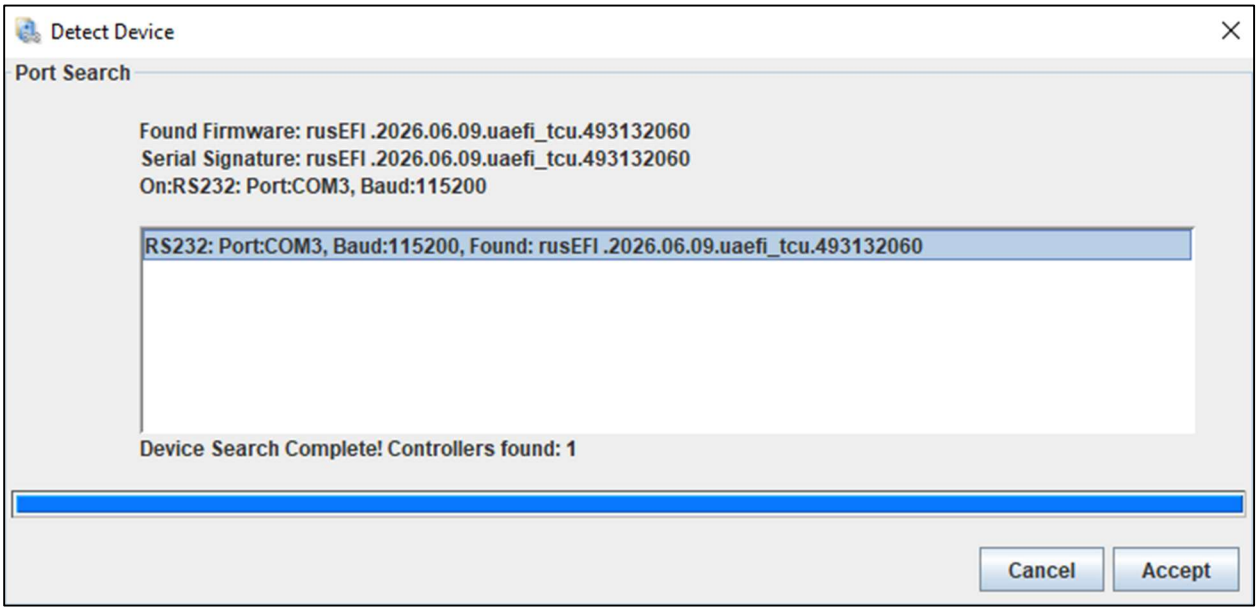


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f. Under “Firmware” select “Detect”.



g. Select the firmware detected and press “Accept”.

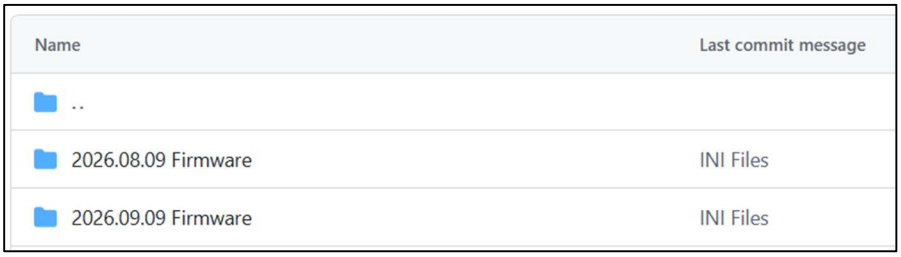


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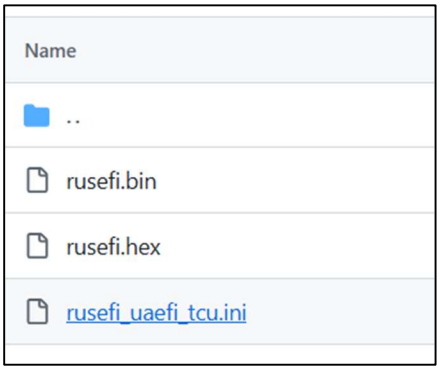
h. What happens if TunerStudio fails to detect or automatically load the .ini file for my firmware?

Navigate to: https://github.com/rmcontrolsystems/PNP-Adapter-Boards/tree/main/TunerStudio/INI%20Definition%20Files

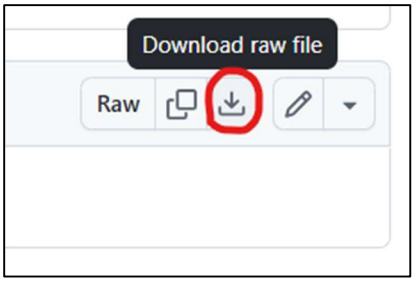
Then navigate to the folder whose firmware date matches the detected firmware date on your ECU



Click on the .ini file

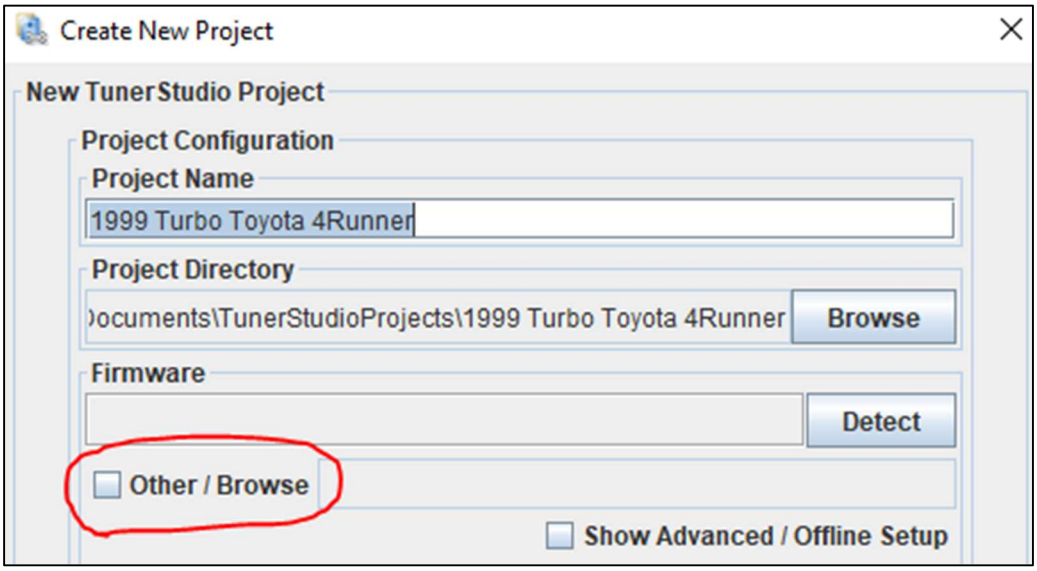


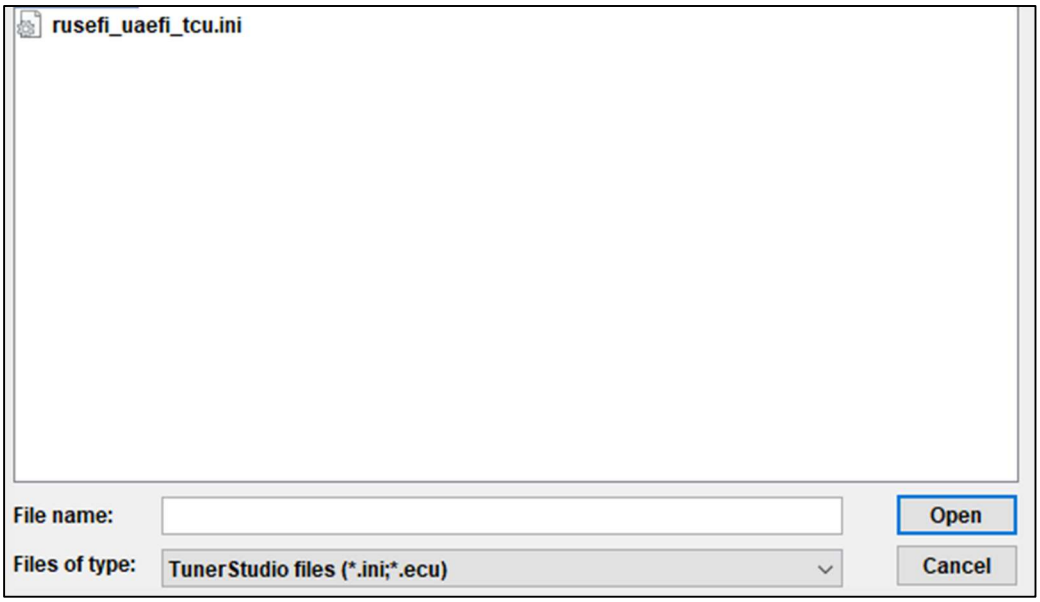
Download the .ini file and save it to your computer



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The .ini file can be loaded manually into TunerStudio by selecting “Other / Browse” and selecting the .ini file that was just downloaded



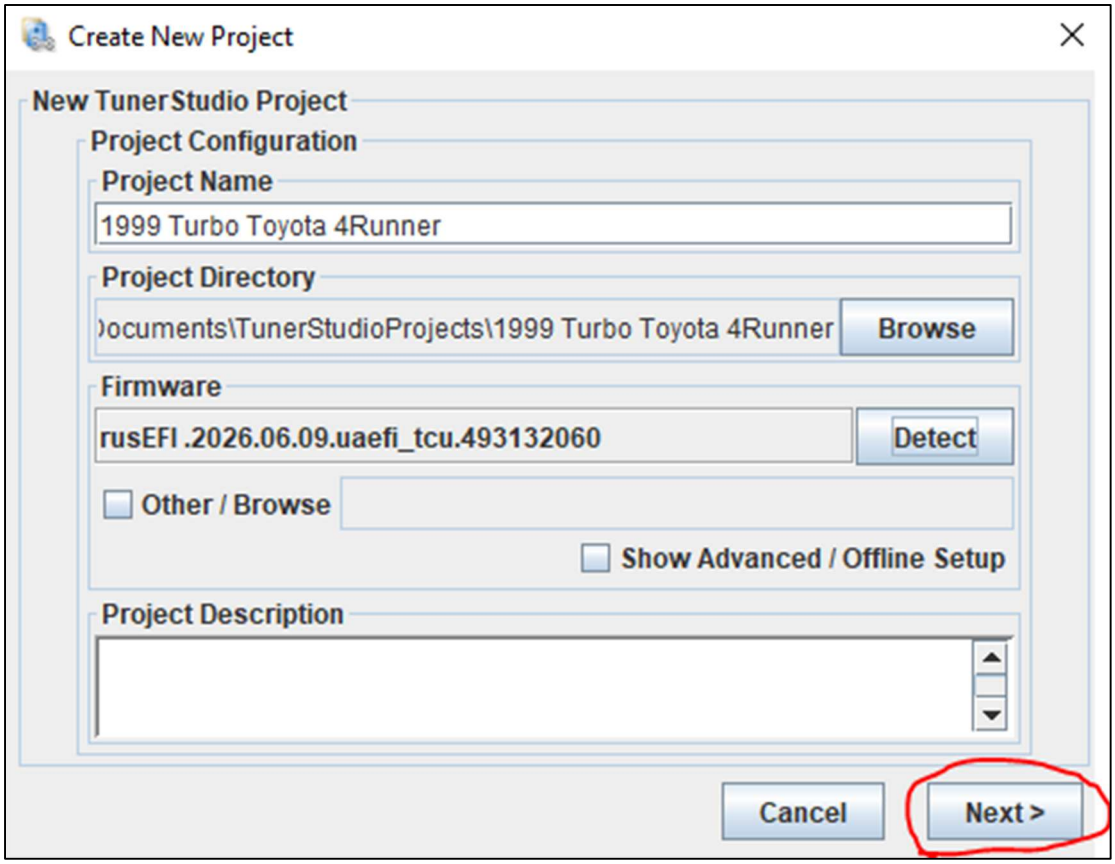


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What is an .ini file?

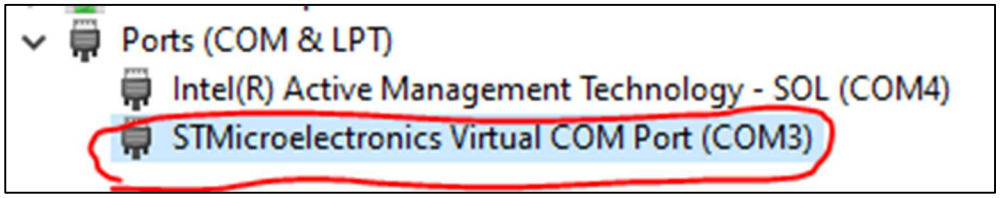
An .ini file tells TunerStudio how to communicate with your ECU and interpret the data it receives. It defines the available settings, gauges, tables, and menus for the specific firmware version running on your ECU. Because different firmware versions may use different features or memory layouts, TunerStudio requires the correct .ini file to communicate properly.

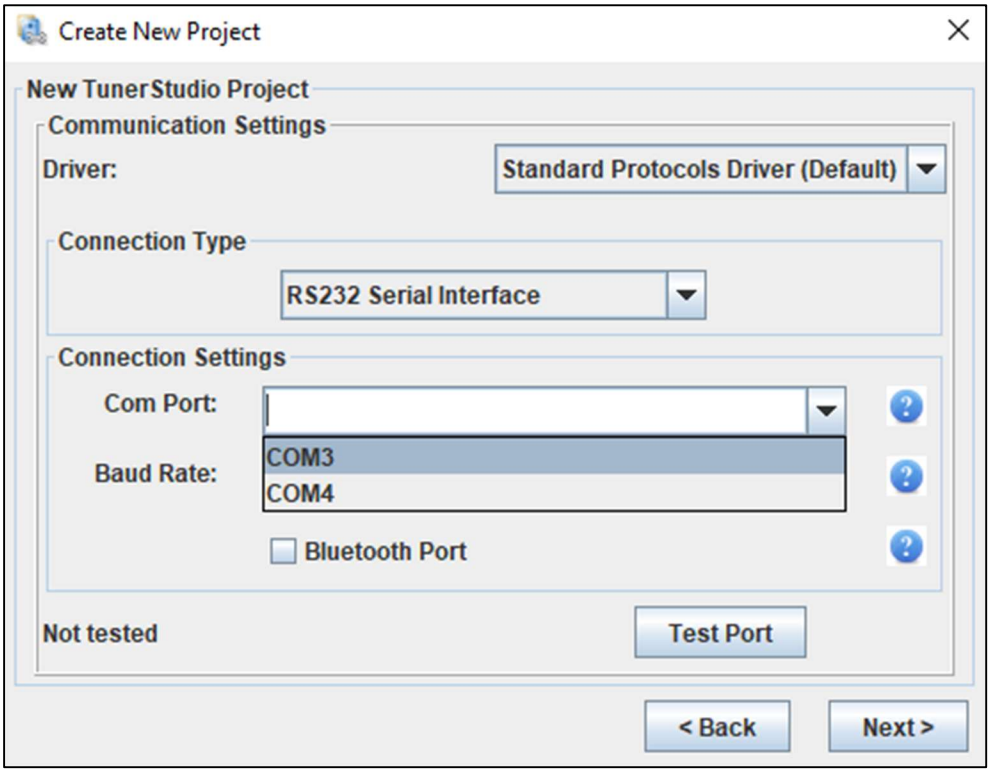
i. Select “Next”



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j. Select the correct port where the ECU is connected to your computer.
To see which port the ECU is connected to, open Device Manager. Under "Ports (COM & LPT)" search for "STMicroelectronics Virtual COM Port" to determine the correct port.



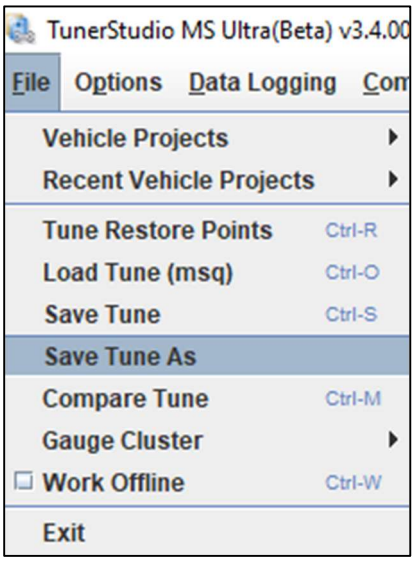


Select the COM port corresponding to "STMicroelectronics Virtual COM Port (COMx)". The COM number may be different on your computer.
Then hit "Next".

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4. <u>Save your current tune!</u>
   Your ECU should now be connected to TunerStudio. Immediately save your current tune before making any changes to it. This is important in case you need to return to the base tune file.

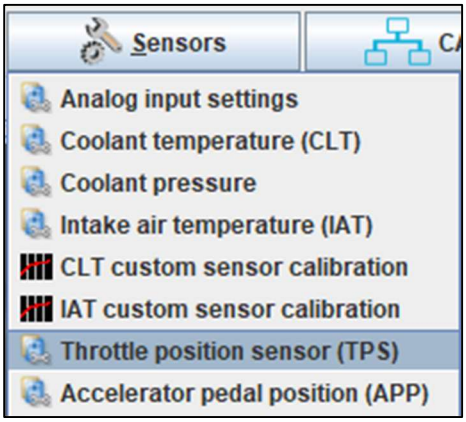
   a. File → Save Tune As…



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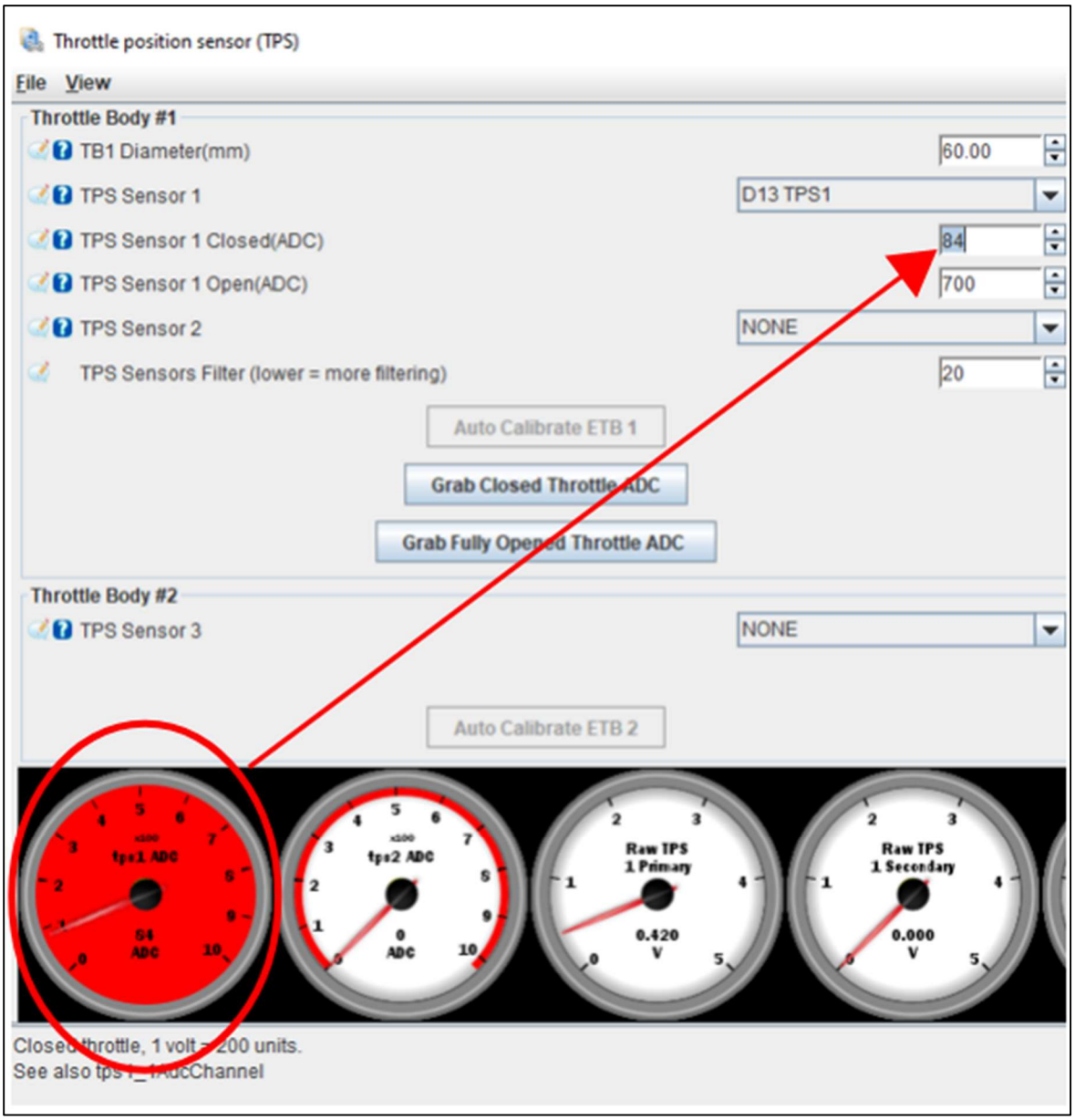
5. <u>Calibrate your Throttle Position Sensor (TPS)</u>
   Once your ECU is connected to Tuner Studio, one of the first things you should do is calibrate your TPS. This is important, particularly for accurate automatic transmission control.

   a. Navigate to…
      Sensors → Throttle Position Sensor (TPS)



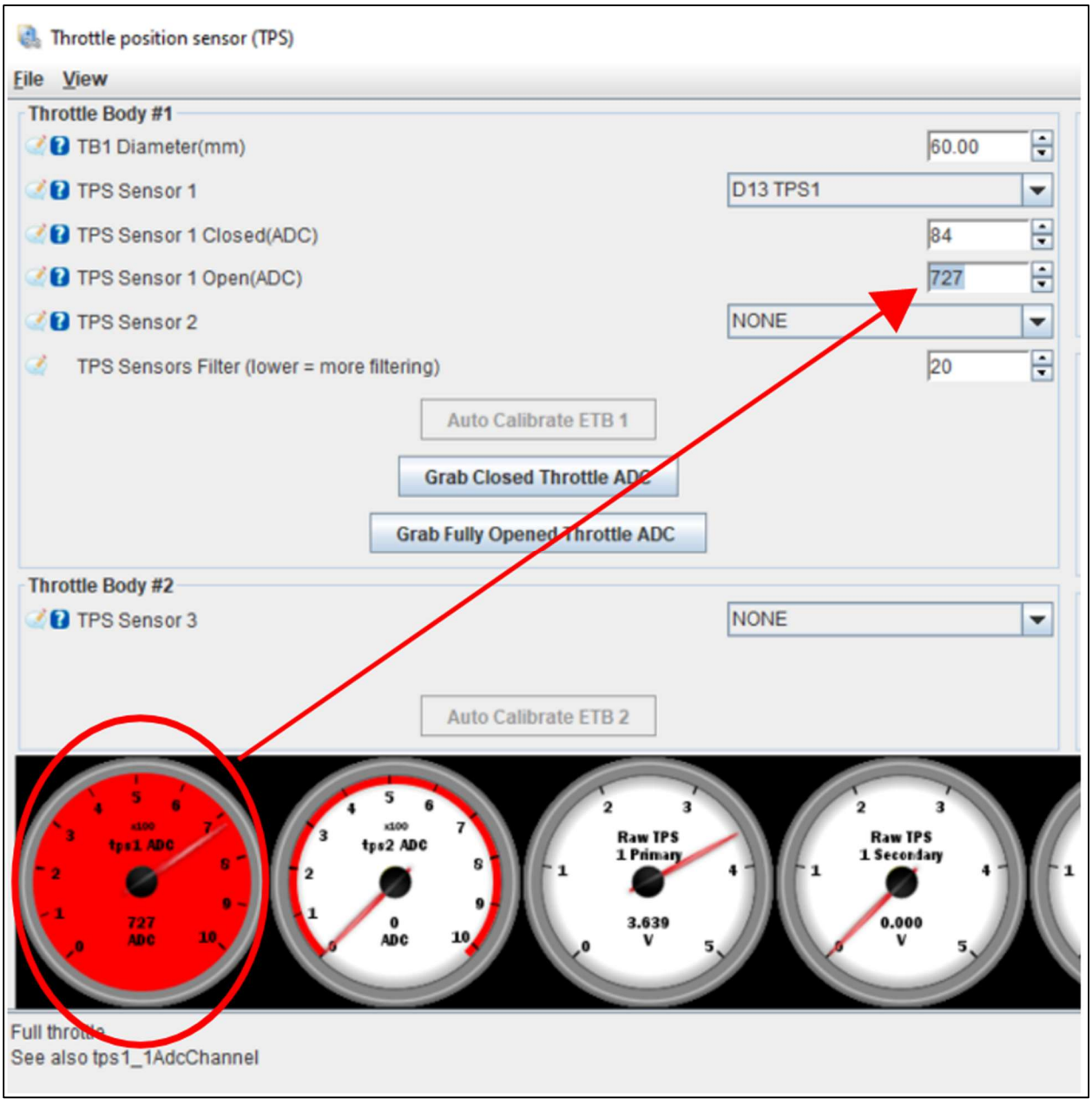
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b. With the throttle completely closed, note the value shown in the ADC gauge. Enter that value into *TPS Sensor 1 Closed/(ADC)*



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c. With the engine off and throttle wide open, note the value shown in the ADC gauge. Enter that value into *TPS Sensor 1 Open/(ADC)*



d. Click “Apply” and then “OK” at the bottom of the menu to save changes and exit.

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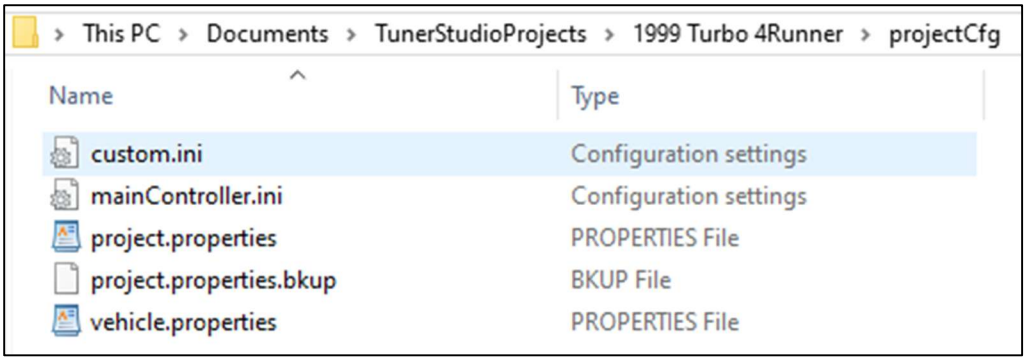
6. Setting Up Custom Dashboards

   TunerStudio uses metric unit by default, which may not be desirable for many users.

   a. Navigate to https://github.com/rmcontrolsystems/PNP-Adapter-Boards/tree/main/TunerStudio/Dashboards and download the `custom.ini` file



   Save the `custom.ini` file into the projectCfg folder inside your TunerStudio project.

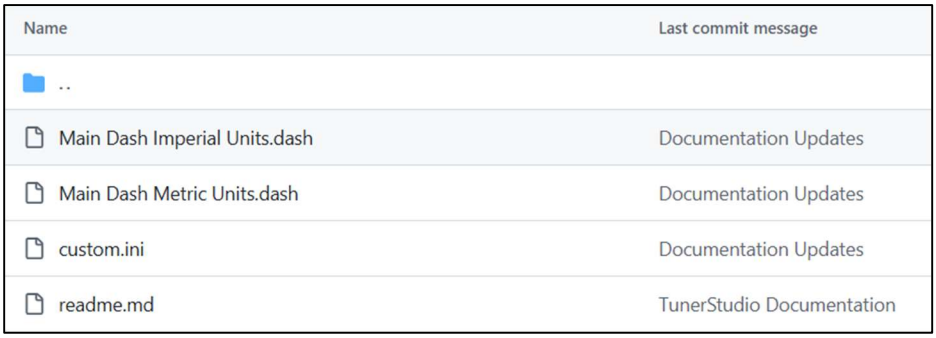


   The typical Windows path is:
   *Documents**TunerStudioProjects*\[Project Name]*projectCfg*

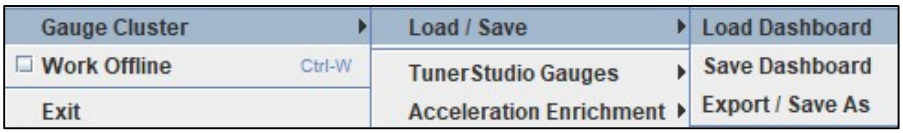
   After the `custom.ini` file has been saved into the projectCfg folder, close and restart TunerStudio.

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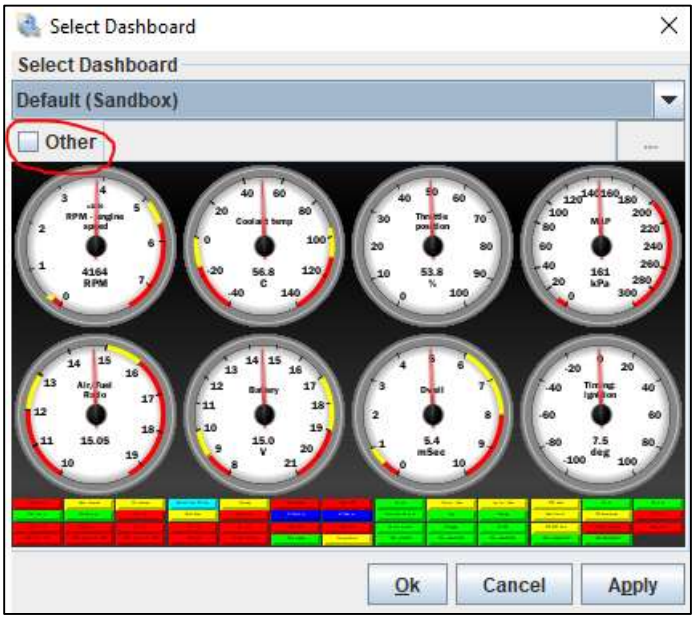
b. Download the desired dashboard (.dash file)



c. In TunerStudio, navigate to **File → Guage Cluster → Load/Save → Load Dashboard**



In the dialog box, select **Other** and then select the downloaded .dash file.



d. Save the dashboard layout if prompted.

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Need help?

Email:
RMControlSystems@gmail.com

Facebook:
RM Control Systems
https://www.facebook.com/profile.php?id=61580687264002

Documentation/Downloads:
https://github.com/rmcontrolsystems/PNP-Adapter-Boards

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